

Requirement understanding document

Functional Requirements

POV:-

A) Admin

B) User

-----Admin-----

1) Member Management

View member

add member

search member

deactivate member

update membership

exception handling :-

Validation for invalid phone number , email

Validation for duplicate phone number , duplicate email

Validation for invalid membership number

2) Book Management

Add book

Add copies

View all books

View available books

Search books

Mark damaged copies

Add category

View categories

3) Borrowing

Borrow validation

Membership rules

Availability checking

Exception handling:-

Handling invalid member id

Handling invalid copy id

Prevent borrowing if fine > 500

4) Returns

Fine calculation (10 per delayed day)

Calculated with normal due date fine with other fines like damaged and lost book fines

Return status update

Exceptions handling of borrow id not found , payment already done , and excess amount payment

5) Reports

Books Currently Borrowed

Overdue Books

Members with Pending Fines

Most borrowed books

Available Books by Category

Member Borrowing History

6) Fine Management

1. View Pending Fines

2. Pay Fine

3. View Fine History

4. Calculate Total Fine

Exception handling in all the modules including invalid fine id , invalid userid , and no fines found

-----USER-----

1. View Profile

Shows user profile

2. View Available Books

To see all the available copies of books

3. Search Books

Can search for books

4. View My Borrowed Books

Can see users borrowed books

5. View Pending Fines

Can see pending fiens

6. View Fine History

All the fine history of a aparticular user

7. Calculate Total Fine

Total fines of the user

8. Exit